

Data Constraints

Constraints are simply rules applied to data columns in a table. They are used to limit the type of data that can go into a table, ensuring the accuracy, reliability, and integrity of the data inside your database.

Here is a quick summary table, followed by a detailed breakdown:

Constraint	Primary Purpose
NOT NULL	Prevents empty values in a column.
UNIQUE	Ensures no duplicate values exist in a column.
PRIMARY KEY	Uniquely identifies each record (row) in a table.
FOREIGN KEY	Connects tables and ensures relationship integrity.
CHECK	Validates data against a specific condition.
DEFAULT	Automatically fills a default value if none is provided.

1. NOT NULL

This constraint enforces a column to not accept **NULL** values. This means that you cannot insert a new record or update a record without adding a value to this specific column.

- **Best Used For:** Essential information like names, phone numbers, or passwords.
- **Example:** `FirstName VARCHAR(50) NOT NULL`

2. UNIQUE

The **UNIQUE** constraint ensures that all values in a column are completely different from one another. No duplicates are allowed.

- **Best Used For:** Email addresses, username IDs, or National Identity Card (NIC) numbers.
- **Example:** `Email VARCHAR(100) UNIQUE`

3. PRIMARY KEY

A **PRIMARY KEY** uniquely identifies each row/record in a database table. It is essentially a combination of a **NOT NULL** and **UNIQUE** constraint. **A table can have only one Primary Key.**

- **Best Used For:** User IDs, Student IDs, or Order Numbers.
- **Example:** `StudentID INT PRIMARY KEY`

4. FOREIGN KEY

A **FOREIGN KEY** is used to link two tables together. It is a field (or collection of fields) in one table that refers to the **PRIMARY KEY** in another table. It prevents actions that would destroy the link between tables.

- **Best Used For:** Linking an "Orders" table to a "Customers" table using the CustomerID.
- **Example:** **FOREIGN KEY (CustomerID) REFERENCES Customers(CustomerID)**

5. CHECK

The **CHECK** constraint ensures that the values entered into a column satisfy a specific condition. If the condition evaluates to false, the record will not be saved.

- **Best Used For:** Age restrictions (e.g., must be over 18) or ensuring a product price is greater than 0.
- **Example:** **Age INT CHECK (Age >= 18)**

6. DEFAULT

The **DEFAULT** constraint sets a default value for a column. This value will be automatically added to all new records if no other value is specified by the user during data insertion.

- **Best Used For:** Setting default statuses (e.g., "Active"), or default countries.
- **Example:** **Country VARCHAR(50) DEFAULT 'Sri Lanka'**

Student Activity: Designing the "Science Club" Table

Scenario: Your school is starting a new Science Club. The club president has asked you to create a database table to store the details of the students who are registering.

Instructions: Write a single CREATE TABLE SQL statement to create a table named ScienceClub. You must apply the correct MySQL constraints based on the following business rules:

1. **MemberID:** Every member must have a unique ID number to identify them. This cannot be left empty. (Needs to be a whole number).
2. **FullName:** We must know the student's name. This field cannot be left blank. (Needs to be text).
3. **Email:** Every student must provide an email address, and no two students can register using the same email. (Needs to be text).
4. **Age:** The club is only for senior students. The student's age must be 16 or older. (Needs to be a whole number).
5. **MembershipStatus:** When a student registers, their status should automatically be recorded as 'Pending' unless specified otherwise. (Needs to be text).